Document Investigation Dossier

Case **DEMO-GENUINE** · generated 06 Nov 2026, 17:37 · analyst: Ayush Kumar

4	0	0	1	LOW
documents	potential manipulation	inconsistencies	observations	review priority

Executive summary

4 documents were examined (Identity Card, Salary Slip, Loan Application Form, Bank Statement). The system recorded 0 potential manipulation hypotheses, 0 detected inconsistencies and 1 observations. Review priority: LOW. No converging signs of alteration and no cross-document conflicts were found. This does not prove authenticity; it means the checks implemented here found nothing to escalate. All findings are indicators for human review. None of them is a determination of fraud.

Status definitions. Observation: a visual or technical characteristic was observed; on its own it is not a problem. **Detected inconsistency**: a verifiable conflict in the data (arithmetic or between documents). **Potential manipulation**: several independent signals converge on the same field. None of these is a finding of fraud; consequential decisions remain with a human reviewer.

7.1 Document overview

ID	File	Type (confidence)	Source	Pages	Status	Review priority	SHA-256 (first 16)
DOC-01	01_identity_card.jpg	Identity Card (99%)	scanned (OCR)	1	investigated	LOW (0.05)	792b4c67dbe4cf16
DOC-02	02_salary_slip.jpg	Salary Slip (99%)	scanned (OCR)	1	investigated	LOW (0.0)	9594c398d3ea5e1d
DOC-03	03_loan_application.jpg	Loan Application Form (99%)	scanned (OCR)	1	investigated	LOW (0.05)	3e2cc339f953a550
DOC-04	04_bank_statement.pdf	Bank Statement (99%)	digital PDF	1	investigated	LOW (0.0)	608403b442f55a56

Full SHA-256 hashes are recorded in the audit trail at ingestion so the analysed files can be matched to the originals (ISO/IEC 27037 practice).

7.2 Extracted information

Every value keeps its page and bounding box (x0, y0, x1, y1 in 150-dpi pixels) so it can be traced to its location in the original document. Confidence is the OCR engine's word confidence; for digital PDFs the text layer is exact (99).

DOC-01 · Identity Card · 01_identity_card.jpg

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	96	1	(373, 265, 554, 286)
Date of birth	14/03/1996	14/03/1996	96	1	(373, 336, 530, 359)
ID number	DEMO-4821-7730-15	DEMO-4821-7730-15	92	1	(373, 406, 662, 426)
Gender	Male	Male	96	1	(373, 475, 435, 496)
Address	21 Lake View Road, Bhopal 462001	21 Lake View Road, Bhopal 462001	96	1	(372, 545, 862, 572)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [373].

DOC-02 · Salary Slip · 02_salary_slip.jpg

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	96	1	(413, 285, 594, 306)
Employee ID	NSW-2291	NSW-2291	92	1	(413, 346, 554, 366)
Pay period	September 2026	September 2026	96	1	(412, 405, 643, 432)
Pay date	30/09/2026	30/09/2026	96	1	(412, 466, 570, 489)
Basic	30,000	30,000	96	1	(472, 611, 559, 633)
Provident fund	3,600	3,600	96	1	(1042, 611, 1113, 633)
HRA	12,000	12,000	96	1	(473, 661, 559, 683)
Professional tax	200	200	96	1	(1042, 661, 1088, 680)
Special allowance	10,000	10,000	96	1	(473, 711, 559, 733)
Income tax	3,000	3,000	96	1	(1042, 711, 1113, 733)
Gross earnings	52,000	52,000	96	1	(412, 786, 506, 810)
Total deductions	6,800	6,800	96	1	(412, 846, 488, 870)
Net pay	45,200	45,200	96	1	(411, 916, 506, 940)
Employer	Nimbus Softworks Pvt Ltd	Nimbus Softworks Pvt Ltd	96	1	(84, 78, 663, 108)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [412, 473, 1042].

DOC-03 · Loan Application Form · 03_loan_application.jpg

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	96	1	(463, 215, 644, 236)
Date of birth	14/03/1996	14/03/1996	96	1	(463, 291, 620, 314)
ID number	DEMO-4821-7730-15	DEMO-4821-7730-15	92	1	(463, 366, 752, 386)
Address	21 Lake View Road, Bhopal 462001	21 Lake View Road, Bhopal 462001	96	1	(462, 440, 952, 467)

Field	Value (normalised)	As read	Conf.	Page	Location
Employer	Nimbus Softworks Pvt Ltd	Nimbus Softworks Pvt Ltd	96	1	(463, 515, 817, 536)
Monthly net income	45,200	45,200	96	1	(461, 591, 556, 615)
Loan amount	300,000	3,00,000	91	1	(462, 666, 583, 690)
Loan tenure	36 months	36 months	95	1	(462, 740, 608, 761)
Application date	05/10/2026	05/10/2026	96	1	(462, 816, 620, 839)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [462].

DOC-04 · Bank Statement · 04_bank_statement.pdf

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	99	1	(395, 204, 537, 236)
Account number	XXXXXXXX5521	XXXXXXXX5521	99	1	(395, 242, 569, 273)
Address	21 Lake View Road, Bhopal 462001	21 Lake View Road, Bhopal 462001	99	1	(395, 279, 761, 311)
Statement period	01/09/2026 to 30/09/2026	01/09/2026 to 30/09/2026	99	1	(395, 317, 657, 348)
Closing balance	50,220	50,220.00	99	1	(395, 637, 497, 669)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [396].

Transactions table (structured)

Date	Description	Debit	Credit	Balance
01/09/2026	Opening Balance	-	-	24,450
03/09/2026	UPI GROCERY MART	3,250	-	21,200
05/09/2026	RENT TRANSFER	15,000	-	6,200
12/09/2026	ELECTRICITY BILL	1,180	-	5,020
30/09/2026	SALARY NIMBUS SOFTWORKS	-	45,200	50,220

7.3 Contextual analysis

Cross-document entity matrix

Each attribute is compared with a reference source (identity attributes: identity card; income: bank statement, as an independent third-party record). ✓ = consistent, ✗ = conflict, REF = reference.

Attribute	Reference	Values found (document: value → result)
Name	Identity Card	DOC-01 Identity Card: **Rohan Verma** → REF DOC-02 Salary Slip: **Rohan Verma** → OK DOC-03 Loan Application Form: **Rohan Verma** → OK DOC-04 Bank Statement: **Rohan Verma** → OK
Date of birth	Identity Card	DOC-01 Identity Card: **14/03/1996** → REF DOC-03 Loan Application Form: **14/03/1996** → OK
ID number	Identity Card	DOC-01 Identity Card: **DEMO-4821-7730-15** → REF DOC-03 Loan Application Form: **DEMO-4821-7730-15** → OK
Address	Identity Card	DOC-01 Identity Card: **21 Lake View Road, Bhopal 462001** → REF DOC-03 Loan Application Form: **21 Lake View Road, Bhopal 462001** → OK DOC-04 Bank Statement: **21 Lake View Road, Bhopal 462001** → OK
Employer	Salary Slip	DOC-02 Salary Slip: **Nimbus Softworks Pvt Ltd** → REF DOC-03 Loan Application Form: **Nimbus Softworks Pvt Ltd** → OK DOC-04 Bank Statement: **SALARY NIMBUS SOFTWORKS** → OK
Monthly net income	Bank Statement	DOC-03 Loan Application Form: **45,200** → OK DOC-02 Salary Slip: **45,200** → OK DOC-04 Bank Statement: **45,200** → REF

Inconsistencies within and across documents

None detected.

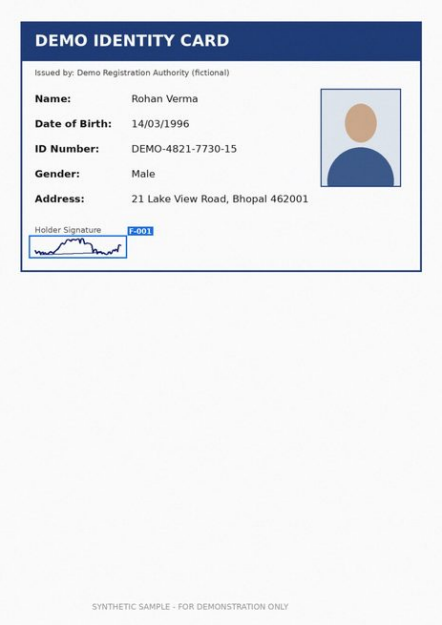
7.4 Visual forensic findings

ID	Finding	Status	Sev.	Conf.	Related
F-001	Signature comparison Identity Card vs Loan Application Form: visually consistent (score 0.96)	Observation	low	50%	-

Annotated pages and forensic maps

Left: findings drawn on the page (red = potential manipulation, orange = inconsistency, blue = observation). Middle: noise-consistency map (bright = abnormally clean). Right: Error Level Analysis.

DOC-01 · Identity Card



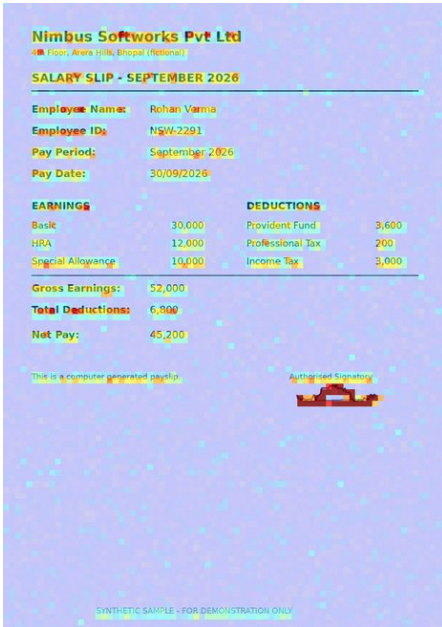




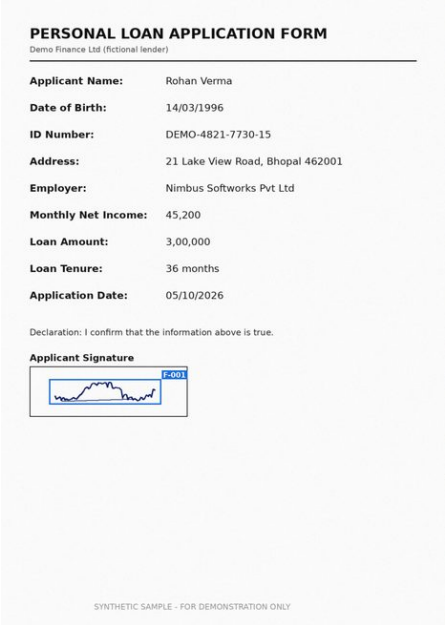
DOC-02 · Salary Slip







DOC-03 · Loan Application Form



PERSONAL LOAN APPLICATION FORM

Demo Finance Ltd (fictional lender)

Applicant Name:	Rohan Verma
Date of Birth:	14/03/1996
ID Number:	DEMO-4821-7730-15
Address:	21 Lake View Road, Bhopal 462001
Employer:	Nimbus Softworks Pvt Ltd
Monthly Net Income:	45,200
Loan Amount:	3,00,000
Loan Tenure:	36 months
Application Date:	05/10/2026

Declaration: I confirm that the information above is true.

Applicant Signature

F-001

SYNTHETIC SAMPLE - FOR DEMONSTRATION ONLY



PERSONAL LOAN APPLICATION FORM

Demo Finance Ltd (fictional lender)

Applicant Name:	Rohan Verma
Date of Birth:	14/03/1996
ID Number:	DEMO-4821-7730-15
Address:	21 Lake View Road, Bhopal 462001
Employer:	Nimbus Softworks Pvt Ltd
Monthly Net Income:	45,200
Loan Amount:	3,00,000
Loan Tenure:	36 months
Application Date:	05/10/2026

Declaration: I confirm that the information above is true.

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SYNTHETIC SAMPLE - FOR DEMONSTRATION ONLY



PERSONAL LOAN APPLICATION FORM

Demo Finance Ltd (fictional lender)

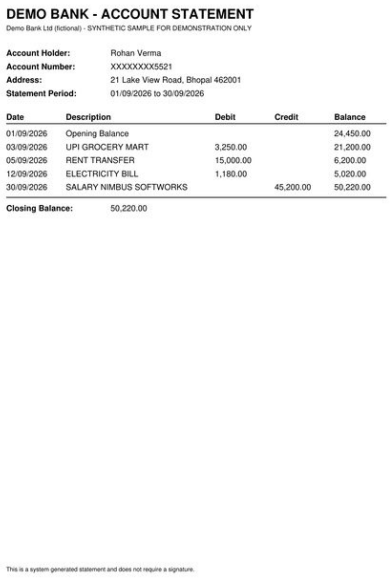
Applicant Name:	Rohan Verma
Date of Birth:	14/03/1996
ID Number:	DEMO-4821-7730-15
Address:	21 Lake View Road, Bhopal 462001
Employer:	Nimbus Softworks Pvt Ltd
Monthly Net Income:	45,200
Loan Amount:	3,00,000
Loan Tenure:	36 months
Application Date:	05/10/2026

Declaration: I confirm that the information above is true.

Applicant Signature

SYNTHETIC SAMPLE - FOR DEMONSTRATION ONLY

DOC-04 · Bank Statement



DEMO BANK - ACCOUNT STATEMENT

Demo Bank Ltd (fictional) - SYNTHETIC SAMPLE FOR DEMONSTRATION ONLY

Account Holder:	Rohan Verma
Account Number:	XXXXXXXX5521
Address:	21 Lake View Road, Bhopal 462001
Statement Period:	01/09/2026 to 30/09/2026

Date	Description	Debit	Credit	Balance
01/09/2026	Opening Balance			24,450.00
03/09/2026	UPI GROCERY MART	3,250.00		21,200.00
05/09/2026	RENT TRANSFER	15,000.00		6,200.00
12/09/2026	ELECTRICITY BILL	1,180.00		5,020.00
30/09/2026	SALARY NIMBUS SOFTWORKS		45,200.00	50,220.00

Closing Balance: 50,220.00

This is a system generated statement and does not require a signature.

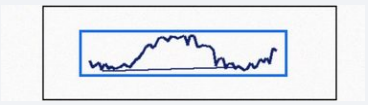
digital PDF: noise analysis not applicable; text-layer font and metadata analysis used instead

7.5 Evidence and traceability

For each finding: the evidence items, their source document, page and bounding box, and an image crop of the original region. Related findings are cross-referenced.

F-001	**Signature comparison Identity Card vs Loan Application Form: visually consistent (score 0.96)**	Observation	LOW · 50%

Shape correlation of the two signature ink masks is 0.96 (1.0 = identical shape). This is a coarse shape heuristic, not handwriting verification: it cannot authenticate a signer and a high score can also mean a signature was copied.

Doc	Page	Location	Field / value	Note	Crop
DOC-01	1	(87, 662, 353, 718)	region	signature A	
DOC-03	1	(138, 1064, 443, 1127)	region	signature B	

Verify: Forensic handwriting examination if the signature matters to the decision. · **Rule:** V4-signature-compare

7.6 Investigation summary

Identified anomalies (ranked)

ID	Anomaly	Status	Conf.

Potential explanations

No converging manipulation hypotheses were raised.

Areas requiring additional verification

Investigation limitations

• Prototype rules and layouts were developed on the synthetic demonstration dataset; new document templates need schema entries.

• Noise-consistency analysis assumes the page was scanned or photographed. Born-digital images have little noise and are analysed with lower sensitivity; a forger who adds matching noise can evade it.

• Glyph-size checks need at least three aligned numeric values in a column.

• Signature detection currently finds coloured (blue) ink only; black-ink signatures may be missed. Signature comparison is a coarse shape heuristic, not handwriting verification.

• Handwritten field recognition is not supported in this version; multilingual OCR is possible through Tesseract language packs but not tested.

• Cross-document reference choices (ID card for identity, bank statement for income) are assumptions; a manipulated reference document weakens the comparison, which is why visual and digital checks run on every document.

• OCR errors can create false inconsistencies; every data finding is scaled by the OCR confidence of the fields it uses, and crops are provided so a reviewer can read the value directly.

• PDF metadata can be altered or legitimately changed; it is used only as supporting evidence.

Appendix A Methodology and technologies

Stage	Technique	Implementation
Ingestion	SHA-256 hashing, page rendering at 150 dpi	Pillow, PyMuPDF
Classification	Keyword-evidence scoring with confidence margin	Python
OCR & extraction	Tesseract LSTM OCR; row/segment layout grouping; label→value pairing with fuzzy label matching; table parsing by column geometry	Tesseract 5, pytesseract, RapidFuzz
Structure	Expected-field schema per document type; missing / duplicate / low-confidence detection; value-column alignment	Python
Visual forensics	Noise-residual consistency (block-wise), Error Level Analysis, glyph-height outliers per value column, coloured-ink signature detection, box detection, signature shape correlation	OpenCV, NumPy
Digital forensics	PDF text-layer font census; metadata (producer vs creator, modification date, incremental saves)	PyMuPDF
Contextual verification	Arithmetic rules (payslip totals, bank running balance), date and age logic	Python rules
Cross-document	Entity matrix with reference sources; fuzzy name/address matching; tolerance-based amount matching; temporal checks	RapidFuzz
Reasoning	Evidence graph keyed by (document, field); convergence of independent signal types; Bayesian-style confidence combination; likely-original estimation; discounting of suspicious corroboration	Python
Narrative (optional)	LLM rewrites the deterministic summary using only structured findings	Claude API (optional)
Evidence & audit	Region crops, annotated pages, hash-chained JSONL audit log, analyst review decisions	Pillow, hashlib

Reference frameworks used as guidance (no formal compliance claimed): ISO/IEC 27037, 27041, 27042; NIST SP 800-86; OWASP ML Security Top 10.

Appendix B Audit trail

Hash-chain integrity: **VERIFIED** - 14 entries verified. Each entry stores the SHA-256 of the previous entry; editing any entry breaks the chain.

#	Time (UTC)	Actor	Event	Details	Hash
1	17:37:30.533	docsentinel	case_opened	case_id=DEMO-GENUINE, files=['01_identity_card.jpg', '02_salary_slip, analyst=Ayush Kumar	bdcb1093fc
2	17:37:30.896	docsentinel	document_ingested	doc_id=DOC-01, filename=01_identity_card.jpg, sha256=792b4c67dbe4cf16d0c116253612254b5e823d4d, media_type=image/jpg, pages=1, source=ocr	39a04c0d23
3	17:37:31.362	docsentinel	document_ingested	doc_id=DOC-02, filename=02_salary_slip.jpg, sha256=9594c398d3ea5e1d071fa70b5880a3773cbde5a6, media_type=image/jpg, pages=1, source=ocr	1cf61271cf
4	17:37:31.793	docsentinel	document_ingested	doc_id=DOC-03, filename=03_loan_application.jpg, sha256=3e2cc339f953a5500b262d1989462a3cb865cc4c, media_type=image/jpg, pages=1, source=ocr	433c130bc8
5	17:37:31.917	docsentinel	document_ingested	doc_id=DOC-04, filename=04_bank_statement.pdf, sha256=608403b442f55a56a36635870deef5b5db9fb90f, media_type=application/pdf, pages=1, source=text-layer	f71fb0a31c
6	17:37:31.917	docsentinel	document_classified	doc_id=DOC-01, doc_type=identity_card, confidence=0.99	d3f8ed8683
7	17:37:31.918	docsentinel	document_classified	doc_id=DOC-02, doc_type=salary_slip, confidence=0.99	5cf43e7a4a
8	17:37:31.918	docsentinel	document_classified	doc_id=DOC-03, doc_type=loan_application, confidence=0.99	c63dbd10f4
9	17:37:31.918	docsentinel	document_classified	doc_id=DOC-04, doc_type=bank_statement, confidence=0.99	2c4540ab09
10	17:37:31.918	docsentinel	fields_extracted	doc_id=DOC-01, n_fields=5, mean_conf=95.2	784db67f94
11	17:37:31.920	docsentinel	fields_extracted	doc_id=DOC-02, n_fields=14, mean_conf=95.7	34d2215fd5

#	Time (UTC)	Actor	Event	Details	Hash
12	17:37:31.920	docsentinel	fields_extracted	doc_id=DOC-03, n_fields=9, mean_conf=94.9	4c6f310c76
13	17:37:31.922	docsentinel	fields_extracted	doc_id=DOC-04, n_fields=5, mean_conf=99.0	4497fd5ad5
14	17:37:33.721	docsentinel	finding_recorded	finding_id=F-001, status=OBSERVATION, severity=low, confidence=0.5, rule=V4-signature-compare	9b11fea225